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Project Overview

For my Individual Project, I selected a **Pet Training School** as my mini world. The purpose of this database is to organize information that a pet training school would need to keep track of while operating its business. This includes pet owners, their pets, trainers, training classes, class enrollments, and payments.

I selected this mini world because a pet training school has several different types of information that are connected to each other, making it a good choice for a relational database. For example, an owner can have a pet, a pet can enroll in a training class, a trainer can teach a class, and an enrollment can have a payment associated with it. Instead of keeping this information in separate files or records, the database connects the information through relationships. The main goal of the project was to design a database that represents these real-world relationships while reducing unnecessary duplication of information and keeping the data organized.

System Design Requirements and Research Process

Before creating the database, I considered what information a pet training school would realistically need to store. I researched the general services and information used by pet training businesses in the area where I live and considered the steps that would occur when an owner signs a pet up for training.

What I found was:

1. An owner provides their contact information.
2. Information about the owner's pet is recorded.
3. Trainers are available with different training specialties.
4. Training classes are created and assigned to trainers.
5. Pets are enrolled into appropriate training classes.

6. Payments are recorded for the enrollments.

From these requirements, I determined that the database needed to store information about six main areas: owners, pets, trainers, training classes, enrollments, and payments.

Challenges and Decision Making

One of the main challenges during this project was determining how the different parts of the pet training school should be separated into tables while still remaining connected.

For example, I needed a way to connect pets with training classes without storing all class information directly inside the Pet table. Creating the Enrollment entity solved this problem because it acts as the connection between a pet and a class.

Another consideration was matching trainers and classes with the different types of pets. Since the database contains dogs, a cat, and a rabbit, I created trainer specialties and classes that were appropriate for those animals. This made me feel that the sample database was more realistic. I also had to make sure that the primary and foreign keys were set up correctly in MySQL Workbench. The EER diagram was useful for visually checking these relationships. Overall doing this for the first time was a challenge as well.

Project Outcomes

The completed database resulted in a functional relational database for a pet training school and is able to organize information about owners, pets, trainers, classes, enrollments, and payments while maintaining relationships between the different types of information.

The project also helped demonstrate the complete database design process, such as designing ER diagrams, implementing the database in MySQL Workbench, inserting sample data, and testing the database using SQL queries. The final database contains six related tables and sample records that show how the system could be used in a pet training school environment.

Conclusion

The Pet Training School Database demonstrates how a relational database can be used to organize information for a real-world business scenario. Through this project, I gained

experience identifying system requirements, designing database entities and relationships, working with primary and foreign keys, creating ER diagrams, implementing tables in MySQL Workbench, inserting data, and retrieving information using SQL. Overall, the final database meets my original goal of creating an organized database that can track pets from their owners through their training classes, trainers, enrollments, and payments.